End-to-end modeling of kilonova light curves and gravitational waveforms from neutron star mergers and possible constraints on nuclear physics parameters

by

Diego Fernando Hernández Porras

Submitted to the Department of Physics
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ABSTRACT

Neutron stars are compact objects whose hydrostatic equilibrium is determined by the equation of state (EOS) of the dense matter in their interiors, which makes them natural laboratories for nuclear physics at densities inaccessible to terrestrial experiments. Although measurements of neutron-star masses, radii, and tidal deformabilities have constrained the pressure of dense matter, there is still no consensus on the composition of the stellar core, in particular on whether a phase transition to deconfined quark matter takes place. Binary neutron-star mergers (BNSMs) probe even higher densities and temperatures and, as demonstrated by the gravitational-wave event GW170817 and its associated kilonova, are a site of rapid neutron-capture (r-process) nucleosynthesis. In these events, the EOS shapes the hydrodynamic evolution of the merger and may therefore leave fingerprints both in the gravitational waveform and in the mass and initial thermodynamic conditions (e.g., density and entropy) of the ejected material. The subsequent nucleosynthesis in the ejecta, however, also depends on the properties of neutron-rich nuclei far from stability, including nuclear masses, β^--decay and neutron-capture rates, and fission properties. Because dense-matter and nuclear-physics uncertainties affect the same observables, disentangling them is essential to determine what current and future multimessenger observations can reveal about dense matter. In this work, we perform a joint analysis of the gravitational-wave and electromagnetic signatures of BNSMs to investigate whether a hadron–quark phase transition leaves an imprint on the gravitational-wave signal and on the ejected material, whose mass and composition determine the radioactive heating that powers the kilonova. To this end, we perform general-relativistic hydrodynamic simulations of an equal-mass BNSM with the Einstein Toolkit, using a hadronic EOS, a hybrid EOS and an alternative hadronic EOS.Post-merger differences can be attributed to the phase transition, while the alternative hadronic EOS allows us to assess whether the combined gravitational-wave and electromagnetic signatures can distinguish a phase transition from a change in the stiffness of hadronic matter.We compare the post-merger gravitational-wave signals and the dynamical ejecta, whose thermodynamic evolution, neglecting neutrino interactions, we post-process with the nuclear reaction network WinNet to obtain r-process abundances and radioactive heating rates, from which we compute kilonova light curves. Finally, we quantify the impact of the phase transition relative to the variation of the hadronic EOS, to uncertainties in

nuclear-physics inputs such as fission-fragment distributions and β^--decay rates, and to numerical resolution.

Thesis supervisor: Neelima Govind Kelkar
Title: Professor of Physics

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Este trabajo respresenta el final de una de las etapas mas bonitas de mi vida. Llevaré con mucho cariño el recuerdo de todos mis amigos a donde me lleve la motivación.

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Chapter 1

Introduction

1.1 Dense matter and the neutron-star equation of state

Neutron stars contain the densest matter known to exist in a stable state. Their central densities reach several times the nuclear saturation density $n_{\rm sat} \simeq 0.15$ fm^{-3} [1]. At densities reached in neutron stars atoms configurations are unlikely and matter adopts a different form. Physicists called this state of matter, in absent of better names, dense matter and has been the protagonist in the study of compact objects. To understand its composition an properties we should begin by the lightest tail of the dense matter. Between densities of $n_{drip} \simeq 2.5^{-4}$ fm^{-3} to $n_{\rm sat}$ neutrons inside nuclei begin to unbound so there are nuclei, electrons and free neutrons. When density reach saturation the difference between inside and outside of a nucleus vanish effectively becoming a continuum of nucleons [2]. At the same time the β equilibrium favors neutrons instead of protons, this leads to a neutron dominated composition. The mechanism from which neutrons drip at such densities and why nature favors neutrons will be addressed in chapter 2.

Neutron stars masses ranges from $\simeq 1.1 - 2M_{\odot}$ with radius from $\simeq 5 - 15\,$km [3]. Their structure is determined by the equation of state (EOS) of this matter through the Tolman-Oppenheimer-Volkoff equations plus rotational corrections. Since the matter inside the star is excepted to be dominated by neutrons, the EOS will inherit the nature predicted from the quantum behavior and interactions of neutrons. Phenomena such as pauli exclusion principle manifest as the degeneracy pressure and repulsive interaction between neutrons through ω meson exchange adds the necessary pressure to prevent the collapse setting up a specific radius for a given mass and rotation. This link between microphysics to astrophysical observables opened a window to study of dense matter through compact object observations. Neutron stars are therefore natural laboratories for strongly interacting matter at densities that cannot be reached in terrestrial experiments.

However the microscopic theory of this matter, quantum chromodynamics (QCD), cannot at present provide the EOS in the relevant density range. At the baryon chemical potentials of neutron-star cores, QCD is non-perturbative, and lattice QCD is hampered by the fermion sign problem at finite baryon density. Controlled calculations exist in the chiral effective field theory limit (up to 2 times nuclear saturation) [4], and perturbative QCD becomes reliable only at densities of order $40\,n_{\rm sat}$, far above those realized in neutron stars [5]. In the intermediate

regime, where neutron-star cores reside, the EOS is described by phenomenological models whose predictions must be confronted with astrophysical observations.

As a consequence, the composition of neutron star cores remains unknown. Matter at several times $n_{\rm sat}$ may consist of nucleons, may contain hyperons, or may undergo a transition to deconfined quark matter, either through a smooth crossover or through a first order hadron-quark phase transition [6]. Heavy ion collisions probe hot and nearly symmetric ($Y_e \simeq 0.5$) matter and cannot access the cold, neutron rich, β-equilibrated regime of neutron stars. Astrophysical observations therefore play a unique role in constraining this region of the QCD phase diagram.

1.2 Binary neutron-star mergers as multimessenger laboratories

Observations of isolated neutron stars have placed significant constraints on the EOS. The existence of pulsars with masses of about $2\,M_\odot$ [7] and the radii inferred by NICER [8] constrain the pressure of dense matter up to roughly $2-3\,n_{\rm sat}$. These quantities, however, constrain the pressure energy density relation rather than giving a full description of the matter. For example, hybrid stars (Neutron stars with quark cores) can have mass-radius relations nearly indistinguishable from those of purely hadronic stars [9].

Binary neutron star mergers (BNSMs) provide complementary information. During the inspiral, the tidal deformability of the stars imprints the gravitational wave (GW) phase. This has been proved in the GW170817 to constrain the tidal deformability $\tilde{\Lambda}$ to values between 80 and 700 [10]. After the merger, the remnant reaches higher densities and temperatures than any isolated neutron star, so the post-merger dynamics are sensitive to the EOS in a regime that isolated stars cannot probe. In addition, the neutron rich material ejected during the merger undergoes rapid neutron capture (r-process) nucleosynthesis, and the radioactive decay of the newly formed nuclei powers a kilonova. The kilonova AT2017gfo, which followed GW170817, confirmed BNSMs as a site of the r-process [11,12].

Extracting EOS information from these signals is, however, far from straightforward. Each step between the EOS and an observable introduces additional uncertainties: numerical resolution and the treatment of neutrino interactions in merger simulations, the properties of neutron-rich nuclei far from stability, and the atomic opacities of the ejecta. Constraints on the EOS derived from the kilonova of GW170817 illustrate this fragility, as they were later shown to depend sensitively on modeling assumptions Moreover, no further neutron-star merger with an electromagnetic counterpart was detected during the fourth LIGO–Virgo–KAGRA observing run, so the interpretation of future events will rely heavily on theoretical predictions. Systematic studies of how uncertainties propagate from the EOS and nuclear physics to the observable signals are therefore required.

1.3 Phase transitions and nuclear-physics uncertainties

The observational signatures of a hadron–quark phase transition depend on the density at which it occurs. If the onset density lies below the central density of the inspiraling stars,

quark cores are already present before the merger and modify the stellar radii, the tidal deformability, and the tidally ejected material. If the onset is reached only in the post-merger remnant, the inspiral is unaffected and the transition manifests itself only after the merger. In the hybrid model considered in this work, [quark matter is already present in the inspiraling stars], so the transition can affect both phases.

The kilonova depends on the EOS only indirectly, through the mass, velocity, and composition of the ejecta. The composition determines the opacity: material with electron fraction $Y_e \lesssim 0.25$ produces lanthanides, which strongly increase the opacity and shift the emission to longer timescales and redder wavelengths The radioactive heating that powers the kilonova, in turn, depends on the decay properties of nuclei far from stability, many of which have not been measured and must be taken from theoretical models

These two sources of uncertainty act on different physical quantities but on the same observable. An EOS-induced change in the ejecta mass and a change in the theoretical β-decay rates can both alter the luminosity and timescale of the kilonova. Likewise, a purely hadronic EOS with a tidal deformability similar to that of a hybrid EOS produces a nearly identical inspiral. A central question is therefore whether nuclear-physics uncertainties can mask the signature of a phase transition in the electromagnetic signal, and whether the combination with the GW signal can break this degeneracy.

1.4 Objectives and research questions

The objective of this thesis is to perform a proof-of-concept joint analysis of the gravitational-wave and kilonova signals of a binary neutron-star merger, in order to assess whether a hadron–quark phase transition leaves signatures that can be distinguished from those of hadronic EOS variations and from nuclear-physics uncertainties. Specifically, we address the following questions:

1. How do the gravitational-wave signal and the dynamical ejecta (mass, velocity, and thermodynamic conditions) of an equal-mass merger differ between hadronic EOSs and an EOS with a first-order hadron–quark phase transition, and are these differences larger than the numerical uncertainty?

2. How do the EOS-induced changes in the nucleosynthesis yields, radioactive heating, and kilonova light curves compare with those produced by different theoretical prescriptions for β-decay rates, and does the joint GW+EM analysis help disentangle the two?

The study is restricted to a single equal-mass binary, three EOSs [names], and the dynamical ejecta; the long-term disk outflows and neutrinos as a direct observable are beyond its scope.

1.5 Structure of the thesis

Chapter 2

Dense Matter and Neutron-Star Structure

What we know about the state of the metter inside a neutron star. What theoretical result we know almost for sure about the structure and what implies the equations to the star deformability, radii and mass.

2.1 Equation of state of cold, β-equilibrated matter

An overview of the matter conditions inside of the neutron star.

2.2 Hadron–quark phase transitions and hybrid equations of state

What a phase transition should be like and why could this happend?

2.3 Thermal effects in the equation of state

If a phase transition is happeing what does that implies to the neutron star?

2.4 Stellar structure: TOV equations and tidal deformability

given the equations for the curvature of a neutron star and a certain equation fo state what should we expect about tidal deformability?

2.5 Observational constraints

Given all the observations done What have we learn about the actual nature of those stars?

Chapter 3

Binary Neutron-Star Mergers

3.1 Merger phases and remnant fate

3.2 Gravitational waves: inspiral tidal effects and post-merger spectrum

3.3 Mass ejection: tidal, shock-driven, and disk-wind ejecta

3.4 Weak interactions and the electron fraction of the ejecta

3.5 Electromagnetic counterparts

Chapter 4

Numerical Relativistic Hydrodynamics

4.1 3+1 decomposition and evolution formulation

4.2 General-relativistic hydrodynamics in conservative form

4.3 The Einstein Toolkit: initial data, mesh refinement, atmosphere

4.4 Gravitational-wave extraction

4.5 Ejecta diagnostics and unbound-matter criteria

Chapter 5

r-Process Nucleosynthesis

5.1 Nuclear statistical equilibrium and freeze-out

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5.3 Reaction-network equations and numerical solution

5.4 Nuclear inputs: masses, decay rates, neutron capture, fission

5.5 Radioactive heating and thermalization

Chapter 6

Kilonova Light Curves

6.1 Li-Paczyński model

6.2 Radiative transfer and the diffusion regime

6.3 Opacities of r-process matter

6.4 Light-curve modeling

Chapter 7

Methodology on the impact of theoretical decay rates across thermodynamics trajectories

7.1 Binary neutron-star simulation

7.2 From simulation to trajectories: ejecta extraction and expansion

7.3 Initial termodinamic trajectories treatment

7.4 Nuclear reaction network setup and theoretical reaction reates

7.5 Modified Li-Paczyński model and radiative transfer simulations

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Chapter 8

Results

8.1 Merger dynamics

8.2 Dynamical-ejecta properties

8.3 Nucleosynthesis and heating rates

8.4 Kilonova light curves

8.5 Relevant nuclei and uncertainties

8.6 Detectability with current and future telescopes

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Chapter 9

Discussion

9.1 Comparison with previous studies

9.2 Limitations and model assumptions

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Chapter 10

Conclusions and Outlook

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Appendix A

Units and conventions

Appendix B

Convergence tests

Appendix C

Network and equation-of-state configuration details

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